

MetalBin S.A.

Manufacturing of Industrial Metallic Containers

EXECUTIVE REMEDIATION AND HARDENING REPORT

Purple Team Assessment — Project Closure IT / Infrastructure Department | May 2026

Classification: CONFIDENTIAL — Restricted Internal Use

1. REMEDIATION — CUPS PRINTER

1.1 Identified Problem

The Debian CUPS printer was configured in Bridged mode (outside the pfSense perimeter), with port 631 accessible without authentication from any device connecting to the network.

1.2 Correction 1 — Move the printer behind the firewall

The main correction is to incorporate the printer into the perimeter protected by pfSense, assigning it to a dedicated network segment.

2.1 Correction 2 — Hardening of cupsd.conf

Edit the CUPS configuration file to replace open access with specific source restrictions: nano /etc/cups/cupsd.conf.

```
nano /etc/cups/cupsd.conf
```

Change the Listen section:

```
# Before:  
Listen 0.0.0.0:631
```

```
# After:  
Listen 192.168.40.10:631
```

Change the Location access section:

```
# Before (vulnerable):  
<Location />  
    Order allow,deny  
    Allow all  
</Location>
```

```
# Despues (correcto):  
<Location />  
    Order allow,deny  
    Allow from 192.168.40.0/24  
    Allow from 192.168.10.10  
    Allow from 192.168.10.30  
</Location>
```

```
<Location /admin>  
    AuthType Basic  
    AuthClass System  
    Require user @SYSTEM  
    Order allow,deny  
    Allow from 192.168.40.1  
</Location>
```

Aplicar cambios:

```
systemctl restart cups  
systemctl status cups
```

2.4 Verification of CUPS Fix

From Kali (192.168.10.100), verify that port 631 is no longer accessible:

```
nmap -p 631,9100 192.168.40.10
```

Expected result: filtered or connection refused from unauthorized IPs.

From Win11 (192.168.10.10), verify that printing works correctly:

```
Open Browser → http://192.168.40.10:631
```

Expected result: CUPS page accessible and functional from authorized IPs.

3. CORRECTIONS IN pfSense RULES

3.1 New PRINT Interface — Firewall Rules

With the printer now in the 192.168.40.0/24 segment, specific rules are required to control its traffic.

Firewall → Rules → PRINT:

#	Action	Source	Destination	Port	Description
1	BLOCK	192.168.40.0/24	192.168.10.0/24	Any	PRINT cannot initiate connections to LAN
2	BLOCK	192.168.40.0/24	192.168.30.0/24	Any	PRINT cannot initiate connections to DMZ
3	BLOCK	192.168.40.0/24	Any	Any	Default deny — everything else from PRINT

3.2 LAN Rules — Controlled Access to the Printer

Add the following rules in Firewall → Rules → LAN to control who can print. Insert them before the Default allow LAN rule:

#	Action	Source	Destination	Port	Description
New	BLOCK	192.168.10.20 (Win7)	192.168.40.0/24	Any	Win7 OT cannot access the printer
New	PASS	192.168.10.10 (Win11)	192.168.40.10	631, 9100	Win11 can print
New	PASS	192.168.10.30 (Zorin)	192.168.40.10	631, 9100	Zorin can print

3.3 Fix — Restrict Access to pfSense WebUI

Currently, any device on the LAN can attempt to access the firewall administration panel. This must be restricted to a single management IP.

En pfSense WebUI → System → Advanced → Admin Access:

- TCP Port: change from 443 to 8443
- This reduces exposure to the standard port

In Firewall → Rules → LAN, replace the current WebUI access rule with:

#	Action	Source	Destination	Port	Description
Mod	PASS	192.168.10.30 (Zorin)	192.168.10.1	8443	Only Zorin can manage pfSense
New	BLOCK	LAN subnets	192.168.10.1	8443	Block WebUI access from the rest of the LAN

3.4 Enable Logging on Critical Rules

In pfSense, edit each important BLOCK rule and enable the “Log” option. This allows viewing all blocked attempts in real time under Status → System Logs → Firewall.

Rules where logging should be enabled:

- Block PRINT to LAN — to detect if the printer attempts internal connections
- Default deny DMZ — to detect lateral movement attempts from the server

4. Implemented Fixes

ID	Vulnerability	Applied Fix	Expected Result
F-001	Exposed CUPS	Move to PRINT segment 192.168.40.0/24 + cupsd.conf restrictions + authentication	CUPS accessible only from authorized IPs
F-002	Open JetDirect	pfSense rules restrict port 9100 to Win11 and Zorin only	Only authorized users can print
F-003	pfSense WebUI	Port changed to 8443	Firewall management restricted

4.1 Key Lesson from the Exercise

A firewall only protects what is inside its perimeter. Any device configured outside that perimeter such as a printer in Bridged mode is a risk regardless of how well the internal firewall is configured.